Husayn El Sharif

Senior Data Scientist • Machine Learning Engineer

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Professional Summary

Data Scientist and ML Engineer with 10+ years of experience developing data-driven solutions across environmental, engineering, and business domains. Ph.D. in Civil Engineering from Georgia Tech with expertise in machine learning, Generative AI, RAG, deep learning, geospatial analytics, and cloud platforms. Google Cloud Professional ML Engineer and Databricks Certified Generative AI Engineer Associate.

Technical Skills

Programming: Python, SQL, R, MATLAB

Machine Learning: Scikit-Learn, TensorFlow, PyTorch, XGBoost, Deep Learning

Data Engineering & MLOps: NumPy, Pandas, PySpark, MLflow, Airflow, Streamlit, Docker, Git

AI Engineering: Prompt Engineering, Fine-Tuning, Agentic RAG, Claude Code, Codex

Cloud & Data Platforms: Google Cloud, AWS, Databricks, Snowflake, Hugging Face

Visualization: Matplotlib, Seaborn, Plotly, Tableau, Power BI

Geospatial Analysis: ArcGIS, QGIS, GeoPandas, Google Earth Engine

Certifications

- Google Cloud Professional Machine Learning Engineer (2026)
- Databricks Certified Generative AI Engineer Associate (2026)

Selected Machine Learning Projects

Remote Sensing & ML for Water Quality Assessment at Lake Lanier
Built a Sentinel-2 and Google Earth Engine ML pipeline to estimate chlorophyll-a concentrations and detect harmful algal blooms, enabling near-real-time monitoring for Gwinnett County water operations. View Project

Agentic RAG Workflow for Building & Fire Code Review
Built a LangChain agentic RAG app with hybrid search for Cobb County building and fire code review, raising faithfulness from 0.26 to 0.97 with 0.77 context precision and 0.72 recall. View Project

Trial-to-Paid Conversion Prediction & MLOps
Built an end-to-end XGBoost pipeline with on-demand and scheduled batch predictions, automated drift monitoring, and CI/CD using MLflow, FastAPI, Docker, AWS, and GitHub Actions. View Project

Smart Meter Risk Segmentation with Unsupervised ML
Built a k-means pipeline on high-frequency AMI data to segment smart meters by load and power-quality behavior, flagging high-risk assets for proactive utility maintenance. View Project

Professional Experience

Senior Data Scientist (Part-Time) Sept 2025 – present
Demand Delta
Marietta, GA

- Developed an XGBoost lead-scoring system using weather, permit, valuation, and geospatial data, projecting a 12% revenue lift (~$450K per 600 leads) and improving precision-recall performance by 3.3x over a heuristic baseline. Added Platt calibration and SHAP for explainable per-lead insights.
- Built an explainable RAG platform to match project requirements with qualified engineering firms, indexing 7K+ public profiles into ~70K retrieval documents using PostgreSQL/pgvector, embeddings, hybrid search, reciprocal rank fusion, and cross-encoder reranking.
- Built an AI-powered lead-enrichment pipeline with Python, Google Gemini, and the Google Places API to surface high-value industrial prospects for an engineering consultancy, replacing manual prospecting with a scored outreach list and accelerating lead qualification.

Senior Data Scientist (Research Engineer) Aug 2012 – present
Georgia Institute of Technology, School of Civil & Environmental Engineering
Atlanta, GA

- Developed bias-correction and spatiotemporal downscaling algorithms for 16 climate models across two emissions scenarios, producing 4-km daily climate projections while preserving temperature-precipitation correlations. Integrated climate, soil, and crop data into ML models to predict crop yield and irrigation demand and quantify long-range agricultural risks across the ACF River Basin.
- Ported and modernized the SAC-SMA hydrologic model from R to Python using OpenAI Codex, adding flexible time steps, units, and calibration objectives. Achieved 100% model fidelity through rigorous testing, enabling high-performance computational workflows to assess projected changes in Georgia's water availability through the year 2100.
- Developed and evaluated an artificial neural network (ANN) to predict daily solar radiation from meteorological data, reducing RMSE by up to 9.7% versus a calibrated empirical model and supporting long-term agricultural forecasting.
- Integrated and evaluated multi-sensor satellite and gridded climate datasets from NASA, JAXA, and NOAA, performing statistical error, bias, and spatial correlation analyses to assess data reliability for predictive modeling across the continental U.S.
- Containerized geospatial ML workflows using Docker, enabling reproducible deployment and execution of environmental monitoring and predictive modeling pipelines.
- Oversaw $400K+ in annual federal and state research funding as Assistant Director of the Georgia Water Resources Institute.
- Mentored graduate researchers in data science, programming, and ML methods, supporting publications and conference presentations.

Engineering Consultant (Part-Time) Aug 2022 – Aug 2025
4Earth, Inc.
Kennesaw, GA

- Built a TensorFlow-based decision algorithm combining neural networks and dynamic programming to select optimal membrane cleaning actions at 10-minute intervals, evaluating 74K+ system states to balance operational costs, equipment wear, and membrane performance.
- Developed and calibrated a data-driven decision support system for wastewater treatment operations, applying sensor time-series analysis and rule-based algorithms to automate membrane cleaning, detect anomalies, and handle edge cases across 32 operating scenarios.
- Supported AWS IoT integration for near-real-time monitoring of 24 operational KPIs at 60-second intervals, enabling data-driven insights into wastewater treatment performance and supporting the development of AI-powered operational systems.
- Developed physics-based and 3D modeling workflows and trained engineers in their adoption, improving design collaboration, reducing physical prototyping iterations, and shortening project delivery time by 20%.

Data Scientist (Graduate Research Assistant) May 2010 – Aug 2012
Florida Atlantic University, Department of Civil, Environmental, & Geomatics Engineering
Boca Raton, FL

- Built and validated ML models to reconstruct missing environmental data, benchmarking 71 methods across 33 years of daily observations from 15 weather stations to improve data accuracy and statistical consistency for hydrologic and climate analyses.

Education

Ph.D.	Civil Engineering	Georgia Institute of Technology	Atlanta, GA	GPA: 3.97	May 2019
M.Sc.	Civil Engineering	Georgia Institute of Technology	Atlanta, GA	GPA: 3.97	May 2016
B.Sc.	Civil Engineering	Florida Atlantic University	Boca Raton, FL	GPA: 3.81	Aug 2011